



NAME: _____

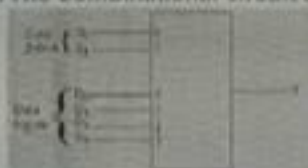
Question 1: Multiple choice Question (30 marks)

1. A combinational circuit that selects one from many inputs are _____
(a) Encoder (b) Decoder (c) Demultiplexer (d) Multiplexer
2. A combinational circuit is one in which the output depends on the _____
(a) Input combination at the time
3. (b) Input combination and the previous output
(c) Input combination at that time and the previous input combination
(d) Present output and the previous output
4. In a multiplexer the output depends on its _____
(a) Data inputs (b) Select inputs (c) Select outputs (d) Enable pin
5. In 1-to-4 multiplexer, if $C_1 = 0$ & $C_2 = 1$, then the output will be _____
(a) Y_0 (b) Y_1 (c) Y_2 (d) Y_3
6. In 1-to-4 multiplexer, if $C_1 = 1$ & $C_2 = 1$, then the output will be _____
(a) Y_0 (b) Y_1 (c) Y_2 (d) Y_3
7. Which of the following is not a sequential circuit?
(a) Flip-flop (b) Counter (c) Shift Register (d) Multiplexer
8. The sequential circuit the output state depend upon _____
(a) Past input state (b) Present input state
(c) Present as well as past input (d) None of the above
9. A 4 bit counter will count
(a) 0 – 3 (b) 0 – 4 (c) 0 – 15 (d) 0 – 16
10. The following truth table represents a _____-gate

X	Y	F
0	0	1
0	1	0
1	0	0
1	1	1

 (a) NOR
 (b) HALF ADDER
 (c) XNOR
 (d) XOR

11. The combinational circuit represents



- (a) Decoder
- (b) Encoder
- (c) Comparator
- (d) Multiplexer

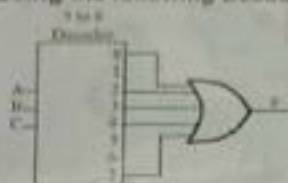
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12. The function represented by the following 4X1 Multiplexer is



- (a) $F = x'yz + x'y'z' + xy$
- (b) $F = xy + xz + x'y'z$
- (c) $F = x'y'z + x'yz' + xy$
- (d) $F = x'y'z + xy'z' + x'y$

13. Using the following Decoder, the simplified SOP for the function F is:



- (a) $F = B'C' + BC + AB'$
- (b) $F = B'C + BC + AB'$
- (c) $F = BC' + BC + AB'$
- (d) $F = B'C' + BC + A'B$

14. To implement $A \oplus B$ using XOR with three inputs, the third input should be

- a) Ground
- b) To one of inputs (A,B)
- c) +5V
- d) Answers b or c

15. Which one of these called universal gate:

- a) NAND only
- b) NAND, NOR
- c) AND, OR, NOT
- d) OR, AND, NOT, NOR, XOR, XNOR, NAND

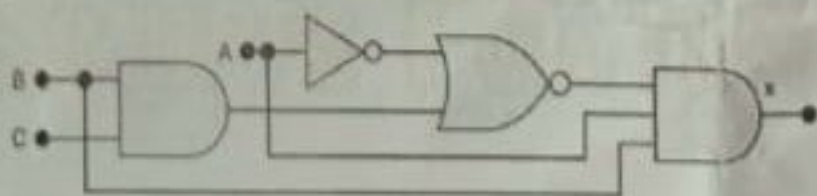
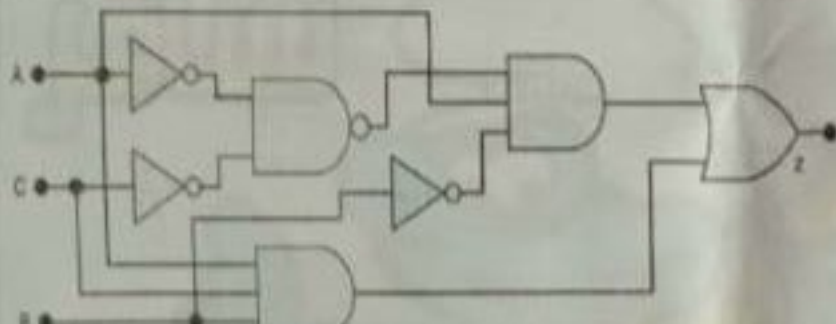
16. What is the output of XOR with two inputs A,B, if A is connected to +5v:

- a) Always zero
- b) Always one
- c) B
- d) B'

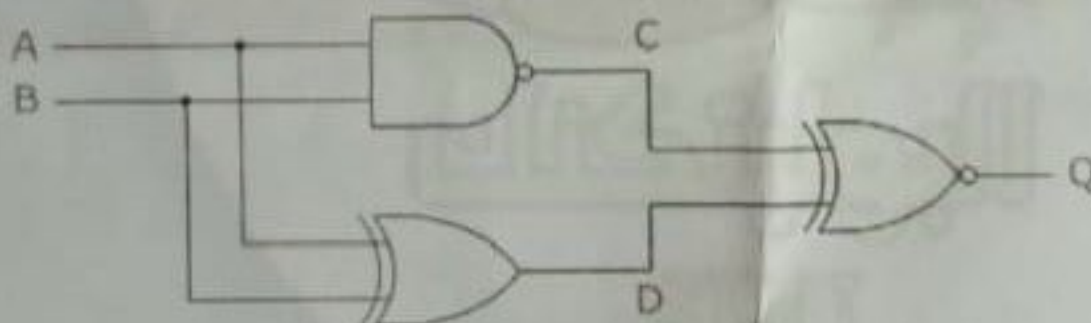
ANSWER:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Question 2: Simplify a logic circuit: (6 marks)

A. 	Final Answer: X =
B. 	Final Answer: Z =
C. $F(A,B,C) = \sum (m_0, m_1, m_2, m_4, m_5, m_6)$	Final Answer: F =

Q4) Construct a Truth Table for the logical functions at point Q in the following circuit and identify a single logic gate that can be used to replace the whole circuit.



Q5) Write a minimal Boolean equation for this 3-bit system: (hint: make the Karnaugh map)

$$F(A, B, C) = \begin{cases} 1 & \text{for } \{0, 1, 2, 4, 6\} \\ 0 & \text{otherwise} \end{cases}$$

Q6) Design a logic circuit with input signal A, control input B, and outputs X and Y to operate as follows:

1. When $B = 1$, output X will follow input A, and output Y will be 0.
2. When $B = 0$, output X will be 0, and output Y will follow input A.